MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEPDOWN, TF5S03ZZ936

(B) Inactive for new design after the date of this specification

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-T-27

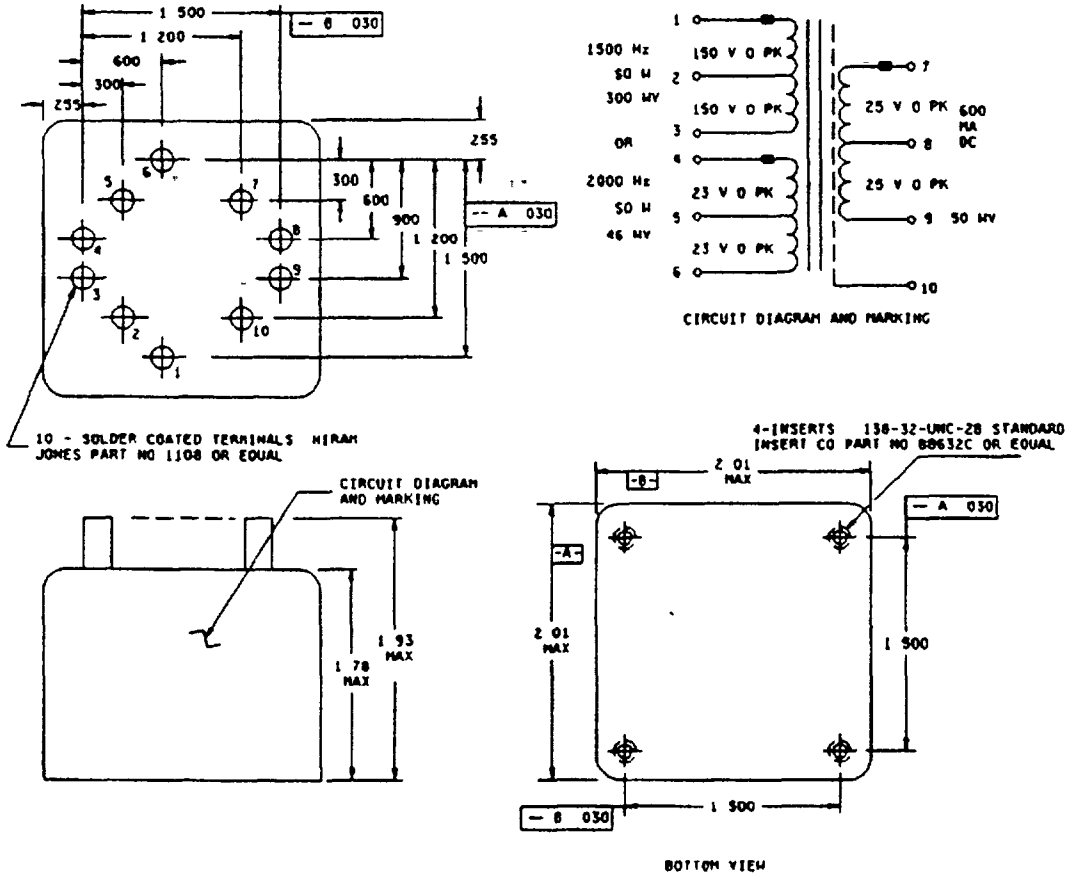


FIGURE 1 Dimensions and configurations

(B) denotes changes

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Inches	mm
030	0 76
138	3 50
255	6 48
300	7 62
600	15 24
900	22 86
1 200	30 48
1 500	38 10
1 78	45 2
1 93	49 0
2 01	51 1

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Marking shall be on side and top of case

FIGURE 1 Dimensions and configurations - Continued

REQUIREMENTS (When numbers in parentheses, i e , (1-3) are used, they indicate the winding and the extreme terminals of the windings)

Electrical ratings

Primary 1 (1-2-3) 150 V, 0 to peak square wave applied to half primary alternately, 1,500 Hz

Primary 2 (4-5-6) 23 V, 0 to peak square wave applied to half primary alternately, 2,000 Hz

Secondary (7-8-9) 25 V, 0 to peak square wave each half secondary, 0 6 ampere dc output from full wave rectifier circuit with choke input filter

Working voltage

(1-3) 300 V

(4-6) 46 V

(7-9) 50 V

DC resistance

(1-3) 60 ohms maximum

(4-6). 1 35 ohms maximum

(7-9) 1 8 ohms maximum

Design and construction

Dimensions and configuration See figure 1

Duty cycle Continuous

Case Epoxy molded

Altitude 10,000 feet maximum

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Terminals Hiram Jones part number 1108 or equivalent (solder coated)

Height 13 inch

Weight $95\frac{3}{4}$ pound maximum

Operating temperature range -55°C to +130°C, maximum

Terminal strength MIL-STD-202, method 211, test condition A, 5 pounds

Dielectric withstanding voltage

(1-2-3) 850 V rms

(4-5-6) 100 V rms

(7-8-9) 500 V rms

Electrical characteristics

No load With 167 V rms, 1500 Hz across (1-2), current in (1-2) shall not exceed 75 mA rms and the power in (1-2) shall not exceed 2 0 watts

With 150 V rms, 1500 Hz on (1-2), voltage across

(2-3) shall be 150 volts +1 percent, -2 percent

(4-6) shall be 45 85 volts +1 percent, -2 percent

(7-9) shall be 52 35 volts +1 percent, -2 percent

Rated load With 150 V, 1,500 Hz across (1-2-3) alternately, or with 23 V, 2,000 Hz across (4-5-6) alternately, and 0 6 ampere dc in (7-9), the voltage across (7-9) shall be 25 V ±5 percent each half secondary

Polarity Additive, with terminals 3 and 4 connected, and terminals 6 and 7 connected

Electrostatic shielding The voltage ratio shall be a minimum of 10 to 1 at 20 kHz

Inductance With 26 V rms, 1,000 Hz, 0 300 A dc applied across (7-8), the inductance (7-8) shall be 10 9 mH ±10 percent

Temperature rise 28°C with 25 5 V rms, 2,000 Hz sine wave across (4-5) at an ambient temperature of 102°C with (7-8) loaded with 0 424 ampere rms

Shock (specified pulse) MIL-STD-202, method 213, test condition letter K

(B) Quality assurance provisions

Qualification inspection Not applicable for this specification sheet

Quality conformance inspection Groups A and B tests of MIL-T-27 shall be applicable

Marking location See figure 1

Part or Identifying Number (PIN) M27/117-01

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CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - OS, SH
Air Force - 17, 80
DLA - ES

User activities
Army - AR, ME
Navy - AS, MC
Air Force - 19

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5950-0815-04)

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